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Research on the motion control strategy of autonomous excavator based on online compensation

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Abstract

The motion control accuracy of the excavator manipulator is the primary guarantee for autonomous excavators to complete work tasks. This paper studies the motion control strategy of the manipulator of a certain autonomous excavator. By establishing a dynamic model of a 3-degree of freedom of excavator manipulator, the gravity term, inertia term, and centripetal force term in the model are equivalent to external disturbances for online compensation, which improves the motion control accuracy of the excavator manipulator. Based on the dynamic model, a control method of the bucket tooth tip trajectory of an autonomous excavator is proposed. The simulation and test results show that the maximum tracking errors of the joint angles of the boom, the bucket, and the bucket are reduced by 43.14, 26.56, and 51.05% respectively, and the average errors of the whole trajectory are reduced by 56.41, 61.33, and 64.26% respectively. The simulation and test results show that the proposed motion control strategy improves the operation accuracy of the excavator, and can effectively improve the operation accuracy and efficiency of the autonomous excavator.

Keywords

Autonomous excavator, dynamic model, excavator manipulator, motion control, online compensation

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Introduction

With the progress of science and technology, Excavators have evolved from simple mechanical hydraulics to a high level of automation and intelligence. The development of unmanned hydraulic excavators is obviously on the rise.[1–3] Curve excavation, straight line excavation, slope excavation, gully slope excavation, and sand and gravel stacking are all extremely repetitive excavation operations carried out by autonomous excavators. The sophisticated nonlinear hydraulic actuator is one of the most difficult aspects of its motion planning and control.[4,5] The control accuracy and working efficiency of the autonomous excavator are jointly determined by the accuracy of the boom hydraulic actuator, the arm hydraulic actuator, the bucket hydraulic actuator, and the rotary hydraulic motor.[6–8] The task determines the work content of the autonomous excavator, and organizes the route planning according to the relevant work objectives. Controlling the hydraulic cylinder through the known planned path results in the telescopic stroke that satisfies the kinematics theory of the excavator manipulator.[9–11]

At present, many scholars have studied the control strategy of excavator manipulators.[12–16] In order to

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realize coordinated control of autonomous excavators, Wang et al.[17] employed the cross-coupling pre-compensation (CCP) method to integrate its compensation with the nonlinear proportional integral controller of each actuator. Lee et al.[18] introduced a contour control algorithm for the leveling operation of the excavator, which improved the leveling accuracy of the excavator. Chang et al.[19] employed a robust control technique to compensate for the nonlinear hydraulic actuator of the excavator manipulator. Hanh et al.[20] created a simplified excavator arm trajectory controller with two degrees of freedom using a fuzzy neural network self-tuning technique. Feng et al.[21] developed an improved genetic algorithm (IGA) to optimize the parameters of the excavator PID controller and obtained an IGA-tuned PID controller. Lee et al.[22] employed data-driven model inversion to execute the offline training model in a supervised way, and used model inversion control to make nonlinear compensation for DX380LC, which substantially improved the tracking accuracy of hydraulic actuators.

The majority of existing research results focus on optimizing the controller design to increase the tracking precision of the excavator manipulator's hydraulic cylinder.[23,24] The control approach they use is to directly adopt the estimated point as the center of gravity of the manipulator, and it is often impossible to precisely determine the actual center of gravity of the manipulator after the bucket of the excavator is loaded.[25] In the field of excavator automation, there are few studies on online identification of excavator mechanical parameters and compensation control of external disturbance compensation of hydraulic cylinders. In this paper, the gravity, centripetal force, and inertial force of the manipulator are online compensated by online identification of the position parameters of the manipulator's center of gravity, which effectively eliminates the influence of external interference during manipulator movement and improves the autonomous excavator's operation accuracy and efficiency. Through the kinematics and dynamics simulation and test of the excavator, the autonomous operation of the excavator is realized. Through the gravity compensation strategy, the influence of gravity is eliminated as much as possible. The inertial force term, centripetal force term, and Coriolis force term generated by the movement of the excavator working device are modeled correspondingly, which increases the accuracy and efficiency of the excavator's autonomous operation. It provides support for the independent operation of excavators in subsequent engineering practice.

Establishment of the online dynamic equation of manipulator

In this section, the manipulator of the autonomous excavator has 3 degrees of freedom. The dynamic model is established as shown in Figure 1. The model mainly includes a boom, arm, bucket device, and its corresponding hydraulic cylinder. Simplify the masses of the boom, arm, and bucket to the centroid of the manipulator, and then the Jacobian matrix of manipulator is calculated.

When solving the dynamic model of the manipulator, the Lagrange technique does not need to consider the internal force of the rod[26,27] where l_1, l_2, and l_3 are the distance from the center of mass of the boom, arm, and bucket to the joint rotation axis, respectively. ϕ_1, ϕ_2, and ϕ_3 is the corresponding angle in Figure 1.The Lagrange function L is defined for the mechanical system of the excavator manipulator as the difference between kinetic energy T and potential energy U of the system.

$$L = T - U \tag{1}$$

Based on the balance principle, the Lagrangian function is established as follows:

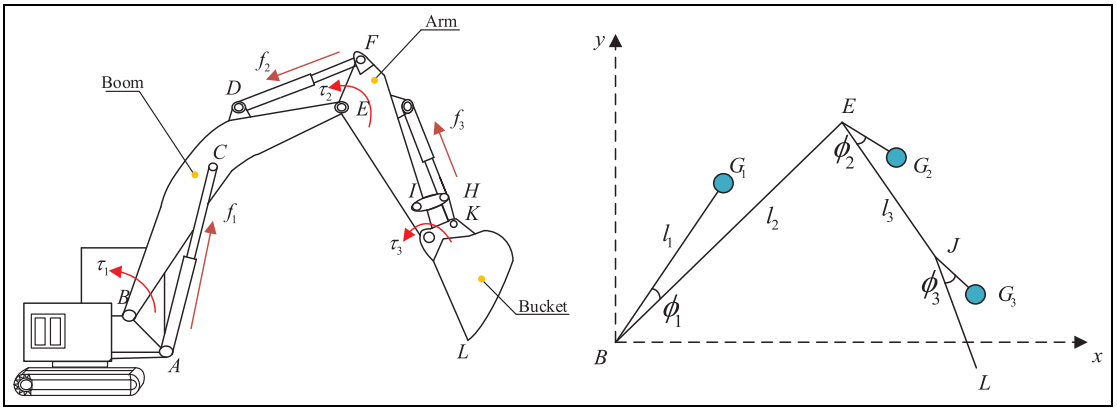


Figure 1. Dynamic model of the working device of the autonomous excavator.

$$\frac{d}{dt}\frac{\partial}{\partial\dot{\theta}_i} - \frac{\partial L}{\partial\theta_i} = \tau_i \ \ (i = 1, 2, \ldots, n) \tag{2}$$

where L is the Lagrangian operator, which represents the difference between the kinetic energy and the potential energy of the system. τ_i is the joint torque.

The kinematic model is shown in Figure 2.

where θ_i and $\dot{\theta}_i$ are joint angle and joint angular velocity respectively. The joint torque of the excavator manipulator is positive in the counterclockwise direction.

The total kinetic energy of the manipulator of the excavator is the sum of the translational energy and the rotational kinetic energy of the center of mass. When the excavator walking device is not in the driving state, only the rotational kinetic energy of the manipulator is considered. The rotational kinetic energy formula of the excavator is as follows:

$$T = \frac{1}{2}\sum_{i}^{3} \dot{\theta}^{T}{}_{i} M_i \theta_i \tag{3}$$

where $\boldsymbol{M}$ is the inertial matrix. The linear velocity and angular velocity of any point on the manipulator can be expressed by the Jacobi matrix and joint velocity as follows:

$$\begin{cases} v_i = J_{vi}(\theta)\dot{\theta} \\ w_i = J_{wi}(\theta)\dot{\theta} \end{cases} \tag{4}$$

where v is the linear velocity of the manipulator and w is the angular velocity of the manipulator. The mass of the boom, arm, and bucket of the manipulator is simplified to the centroid to calculate the Jacobi matrix of the manipulator; the centroid positions of the boom, arm, and bucket are as follows:

$$\begin{aligned} G_1 &= \begin{bmatrix} l_1 \cdot \cos(\theta_1 + \phi_1) \\ l_1 \cdot \sin(\theta_1 + \phi_1) \\ 0 \end{bmatrix} \\ G_2 &= \begin{bmatrix} a_1\cos\theta_1 + l_2\cos(\theta_1 + \theta_2 + \phi_2) \\ a_1\sin\theta_1 + l_2\sin(\theta_1 + \theta_2 + \phi_2) \\ 0 \end{bmatrix} \\ G_3 &= \begin{bmatrix} a_1\cos\theta_1 + a_2\cos(\theta_1 + \theta_2) + l_3\cos(\theta_1 + \theta_2 + \theta_3 + \phi_3) \\ a_1\sin\theta_1 + a_2\sin(\theta_1 + \theta_2) + l_3\cos(\theta_1 + \theta_2 + \theta_3 + \phi_3) \\ 0 \end{bmatrix} \end{aligned} \tag{5}$$

The Jacobian matrix calculation formula of the excavator's manipulator is:

$$J^{c}_{ki} = \frac{\partial\tilde{x}_c(k)}{\partial\theta_i}(k, i = 1, \ldots, n) \tag{6}$$

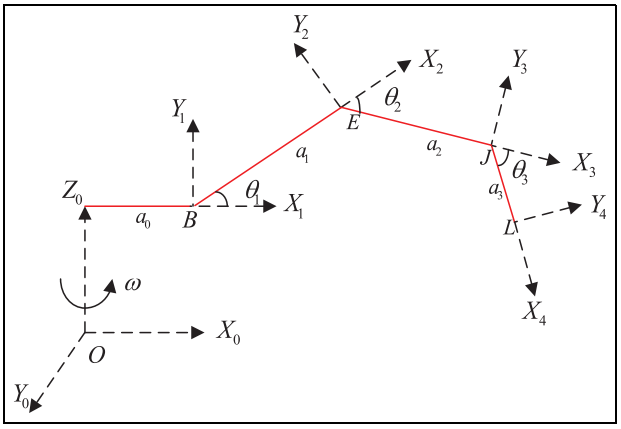


Figure 2. Kinematic model of working device.

where $\tilde{x}_c$ is the posture of the c-bar. The linear velocity and angular velocity at the center of mass of the boom, arm, and bucket can be expressed as:

$$\begin{bmatrix} v \\ w \end{bmatrix} = J \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \end{bmatrix} \tag{7}$$

The specific formula is shown in Appendix A (A-1), and the joint torque of the manipulator can be expressed as follows:

$$\tau = M(\theta)\ddot{\theta} + c(\theta, \dot{\theta})\dot{\theta} + G(\theta) \tag{8}$$

where $\boldsymbol{\tau}$ is the joint moment(N · m),$\boldsymbol{M} \in \mathbf{R}^{3\times3}$ is the inertia matrix of rigid body(kg · m^2),$\boldsymbol{c} \in \mathrm{R}^{3\times3}$ is centripetal force matrix(N · m),$\boldsymbol{G} \in \mathbf{R}^{3\times3}$ is the gravity matrix(N · m).

Related parameter identification and online dynamic compensation

Online gravity compensation and related parameter identification

The weight of the excavator's manipulator is large. During the operation, the dynamic characteristics of the boom lifting, the arm, and the bucket are greatly affected. If the online compensation calculation of the excavator boom, arm, and bucket is carried out, the parameters such as τ_1, τ_2, τ_3, m_1, m_2, m_3, l_1, l_2, l_3, ϕ_1, ϕ_2,ϕ_3 need to be obtained. This paper completes the calculation of kinetic parameters according to the following steps. The order of implementation is as follows.

Firstly, in the test, the pressure sensor is used to measure the driving force f_1, f_2, f_3 of the hydraulic cylinder under different attitudes and no-load conditions, as well as the displacement y_1, y_2, y_3 of the boom, arm, and

bucket hydraulic cylinder at the corresponding time, and the joint angles θ_1, θ_2, θ_3 at this time are obtained by using the forward kinematics solution. At this time, the driving force generated by the change of the hydraulic cylinder is the Coulomb friction, the viscous friction, and the external force along the direction of the hydraulic cylinder caused by self-gravity. Since the influence of the first two is far less than the influence of self-weight, it is assumed that the combined external force is approximately equal to the external force generated by self-weight, that is f_{g1}, f_{g2}, f_{g3}. In this instantaneous state, the driving force of the excavator manipulator is:

$$\begin{cases} f_1 = f_{g1} \\ f_2 = f_{g2} \\ f_3 = f_{g3} \end{cases} \tag{9}$$

At this time, the dynamic equation of the three joint torques $\tau' = [\tau'_1 \quad \tau'_2 \quad \tau'_3]^{\mathrm{T}}$ of the manipulator generated by gravity is shown as follows:

$$\begin{bmatrix} \tau'_1 \\ \tau'_2 \\ \tau'_3 \end{bmatrix} = \begin{bmatrix} m_1 l_1 c_1 + m_2 a_1 + m_3 a_1 & m_2 l_2 c_2 + m_3 a_2 & m_3 l_3 c_3 \\ 0 & m_2 l_2 c_2 + m_3 a_2 & m_3 l_3 c_3 \\ 0 & 0 & m_3 l_3 c_3 \end{bmatrix} \begin{bmatrix} c\theta_1 \\ c\theta_{12} \\ c\theta_{123} \end{bmatrix} g - \begin{bmatrix} m_1 l_1 s_1 & m_2 l_2 s_2 & m_3 l_3 s_3 \\ 0 & m_2 l_2 s_2 & m_3 l_3 s_3 \\ 0 & 0 & m_3 l_3 s_3 \end{bmatrix} \begin{bmatrix} s\theta_1 \\ s\theta_{12} \\ s\theta_{123} \end{bmatrix} g \tag{10}$$

where $c_1 = \cos\phi_1$, $c_2 = \cos\phi_2$, $c_3 = \cos\phi_3$, $s_1 = \sin\phi_1$, $s_2 = \sin\phi_2$, $s_3 = \sin\phi_3$, $c\theta_1 = \cos\theta_1$, $c\theta_{12} = \cos(\theta_1 + \theta_2)$, $c\theta_{123} = \cos(\theta_1 + \theta_2 + \theta_3)$, $s\theta_1 = \sin\theta_1$, $s\theta_{12} = \sin(\theta_1 + \theta_2)$, $s\theta_{123} = \sin(\theta_1 + \theta_2 + \theta_3)$. $[\tau'_1 \quad \tau'_2 \quad \tau'_3]^{\mathrm{T}}$ is the driving torque acting on the boom, arm and bucket only under the action of gravity, m_1, m_2, m_3 are the mass of the boom, arm and bucket respectively.

The driving force of a hydraulic cylinder is the key to generate driving torque. The distance from the force to the shaft is the intermediate medium for the conversion of driving force and torque.

According to Figures 3–5. the relationship between the driving force arm or equivalent length of each cylinder and the length of the cylinder body is obtained by geometric method.

The driving arm E_1 of the boom hydraulic cylinder is:

$$\begin{cases} E_1 = BC \cdot \sin\angle BCA \\ \angle BCA = \arccos\dfrac{BC^2 + \lambda_1^2 - AB^2}{2 \cdot BC \cdot \lambda_1} \end{cases} \tag{11}$$

The driving arm E_2 of the arm hydraulic cylinder is:

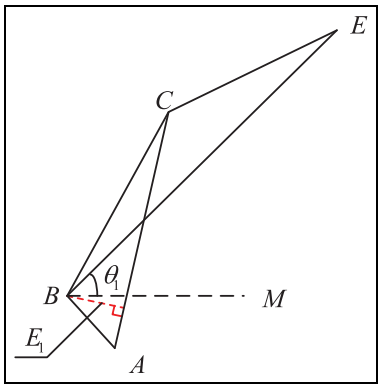


Figure 3. Structural analysis diagram of boom device.

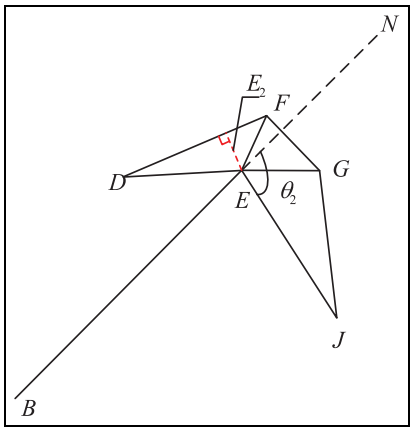


Figure 4. Structural analysis diagram of arm device.

$$\begin{cases} E_2 = \sin\angle FDE \\ \angle FDE = \arccos\dfrac{\lambda_2^2 + DE^2 - EF^2}{2 \cdot DE \cdot \lambda_2} \end{cases} \tag{12}$$

The driving arm E_3 of the bucket hydraulic cylinder is:

$$\begin{cases} l_a = GI \cdot \sin\angle HGI \\ \angle HGI = \arccos\left(\frac{GI^2 + \lambda_3^2 - HI^2}{2 \cdot GI \cdot \lambda_3}\right) \end{cases}$$

$$\begin{cases} l_b = HI \cdot \sin\left(\arccos\frac{HK^2 + HI^2 - IK^2}{2 \cdot HK \cdot HI}\right) \\ IK^2 = IJ^2 + JK^2 - 2IJ \cdot JK\cos(\pi - \theta_3 - \angle EJI - \angle KJL) \end{cases}$$

$$\begin{cases} l_c = JK \cdot \sin\left(\arccos\frac{HK^2 + JK^2 - HJ^2}{2 \cdot HK \cdot JK}\right) \\ HJ^2 = HI^2 + IJ^2 - 2HI \cdot IJ\cos\angle HIJ \\ \angle HIJ = \angle KIH + \angle KIJ \\ \angle KIH = \arccos\frac{HI^2 + IK^2 - HK^2}{2 \cdot HI \cdot IK} \\ \angle KIJ = \arccos\frac{IJ^2 + IK^2 - JK^2}{2 \cdot IJ \cdot IK} \\ IK^2 = IJ^2 + JK^2 - 2 \cdot IJ \cdot JK\cos\angle IJK \\ \angle IJK = \pi - \angle EJI - \angle KJL - \theta_3 \end{cases} \tag{13}$$

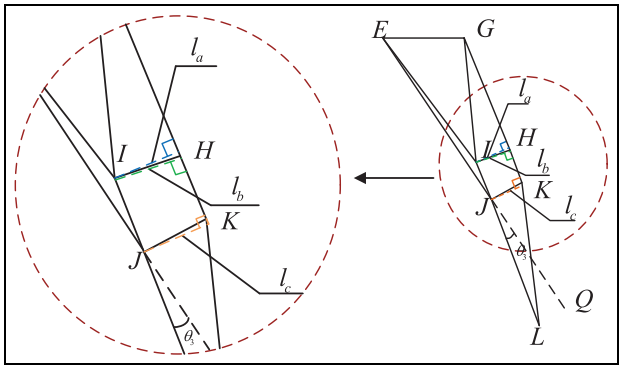


Figure 5. Structural analysis diagram of bucket device.

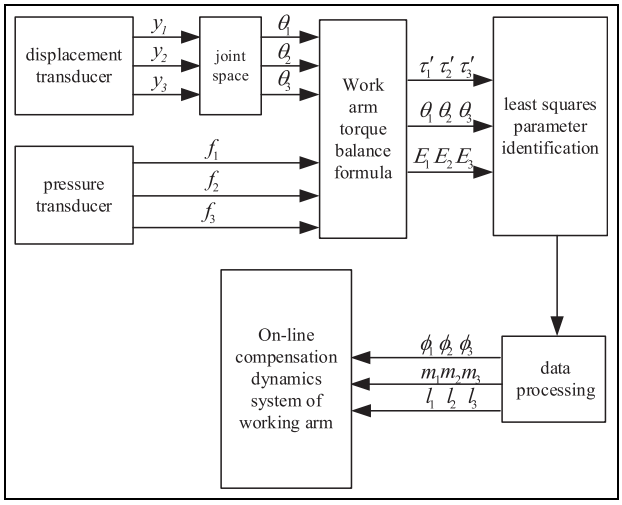


Figure 6. Derivation process diagram of relevant parameters.

The above $\lambda_1, \lambda_2, \lambda_3$ is the total length of the hydraulic cylinder and telescopic rod. where $F_{\text{bucket}} \times l_a = F_{\text{HK}} \times l_b$. The torque $\tau_3 = F_{KH} \times l_c$ of the KH rod to the J point is taken, and the equivalent arm E_3 of the bucket is $\frac{l_a \times l_c}{l_b}$.

According to θ_1, θ_2, θ_3 derived from inverse kinematics, the external forces f_{g1}, f_{g2}, f_{g3} caused by self-weight are calculated as:

$$\begin{cases} f_{g1} = \tau'_1/E_1 \\ f_{g2} = \tau'_2/E_2 \\ f_{g3} = \tau'_3/E_3 \end{cases} \tag{14}$$

Secondly, The joint angles θ_1,θ_2,θ_3 and driving torque τ'_1, τ'_2, τ'_3 are calculated by the displacement y_1, y_2, y_3 and driving force f_1, f_2, f_3 of the boom, arm and bucket hydraulic cylinder obtained by the above steps. The parameters in equation (10) are derived and identified by the least square method. The identification process is as follows (Figure 6):

All the parameters in the $R^{3\times3}$ matrix of formula (10) can be derived. The obtained matrix is processed to obtain the required parameters m_1, m_2, m_3, l_1, l_2, l_3, and so on. The online gravity compensation torque can be obtained by substituting the parameter identification results into Formula (10).

Online centripetal force term and inertial force term compensation

The inertia force and centripetal force created by the enormous weight of the manipulator will affect the motion control precision of the excavator operation, so the centripetal force and inertia force must be compensated online.

The inertial force term is:

$$M(\theta) = \sum_{i=1}^{N} (\mathrm{m}_i J_v^{iT} J_v^i + I_i J_w^{iT} J_w^i) = \begin{bmatrix} M_{11} & M_{12} & M_{13} \\ M_{21} & M_{22} & M_{23} \\ M_{31} & M_{32} & M_{33} \end{bmatrix} \tag{15}$$

The centripetal force term is:

$$c(\theta, \dot{\theta}) = C_{ijk} = \frac{1}{2} \sum_{jk} \left(\frac{\partial M_{ij}}{\partial \theta_k} + \frac{\partial M_{ik}}{\partial \theta_j} - \frac{\partial M_{jk}}{\partial \theta_i} \right) \dot{\theta}_j \dot{\theta}_k \tag{16}$$

The relevant parameters identified in the previous section are substituted into Formulas (15) and (16) to obtain the specific inertial force term and centripetal force term. The formulas are detailed in Appendix A (A-2, A-4).

Combined with the previous section, the online compensation model of gravity, centripetal force, and inertial force can be obtained. The driving force of the boom, arm, and bucket hydraulic cylinder measured by the force sensor during operation is f_1, f_2, f_3. The driving force after the external interference force f_{e1}, f_{e2}, f_{e3} in the direction of the compensated hydraulic cylinder can be used as the driving force f_{R1}, f_{R2}, f_{R3} of the hydraulic cylinder under actual conditions, which is:

$$\begin{cases} f_{R1} = f_1 - f_{e1} \\ f_{R2} = f_2 - f_{e2} \\ f_{R3} = f_3 - f_{e3} \end{cases} \tag{17}$$

Manipulator hydraulic system

Valve-controlled asymmetric cylinder

The boom hydraulic cylinder, arm hydraulic cylinder, and bucket hydraulic cylinder are the three hydraulic cylinders utilized in the excavator manipulator. The expansion and contraction speed of the hydraulic cylinder is determined by the valve opening of the electro-

hydraulic proportional valve in this system. The relationship between the control signal and the expansion amount of the hydraulic cylinder can be determined. Figure 7 depicts a valve-controlled asymmetric hydraulic cylinder system.

The servo amplifier in the controller converts the voltage signal to a current signal. This process is a purely proportional phase, and the transfer function is expressed as:

$$G_1 = \frac{I(s)}{U(s)} = k_a \tag{18}$$

where I is the output current, U is the input voltage, k_a is the servo amplifier amplification factor.

In this section, the flow characteristics of the servo valve are simplified as a second-order oscillation link, which is expressed as:

$$\frac{d^2x_v}{dt^2} + 2\zeta w_m \frac{dx_v}{dt} + w_m^2 x_v = k_v k_a w_m^2 u \tag{19}$$

where x_v is the slide valve displacement of the main valve, w_m is the natural frequency of the servo valve, u is the input servo valve voltage signal, and k_v is the servo valve flow gain.

The load flow equation, hydraulic cylinder flow equation, and hydraulic cylinder force balance equation are all included in the valve-controlled asymmetric hydraulic cylinder.

The linear flow equation of the servo valve can be written as:

$$Q_L = K_q x_v - K_c p_L \tag{20}$$

where Q_L is the load flow, P_L is the load pressure, K_q is the flow gain, and K_c is the flow pressure coefficient.

The load flow formula of the hydraulic cylinder is as:

$$Q_L = C_{tp} p_L + \frac{V_e}{4\beta_e}\frac{dp_L}{dt} + A_{me} v \tag{21}$$

where v represents the piston speed, the equivalent leakage coefficient of the C_{tp} system, V_e represents the entire volume in the hydraulic cylinder cavity, β_e represents the effective bulk modulus of the system., and A_{me} represents the effective cross-sectional area of the piston.

The motion equation of the piston is as:

$$m\frac{d^2y}{dt^2} + B_p\frac{dy}{dt} = A_1P_1 - A_2P_2 + F \tag{22}$$

where A_1 and A_2 are the piston area of the rodless cavity and the rod cavity respectively, K is the load stiffness coefficient, y is the displacement of the hydraulic cylinder, P_1 and P_2 are the rod cavity pressure and the rodless cavity pressure of the hydraulic cylinder, respectively, F is an external load and B_p is the load damping coefficient.

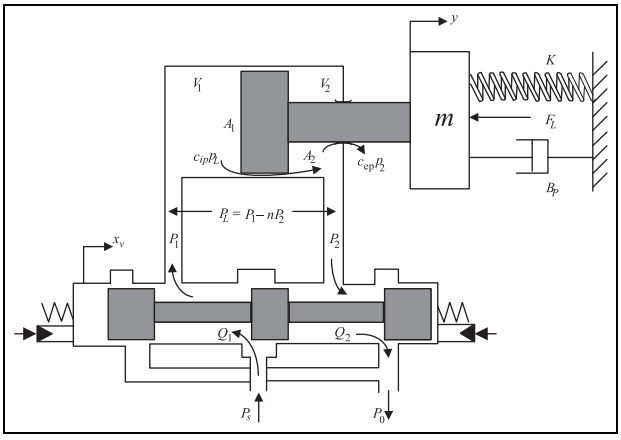


Figure 7. Control diagram of valve-controlled asymmetric hydraulic cylinder.

The Laplace transform of equations (20), (21), and (22) is performed without taking leakage into account. The following is the transfer function of the input spool displacement to the piston displacement of the hydraulic cylinder:

$$Y(s) = \frac{\frac{K_q}{A_{me}}X_v(s) - \frac{K_{ce}}{A_{me}^2}(1 + \frac{V_e}{4\beta_e K_{ce}}s)F(s)}{s(\frac{s^2}{w_n^2} + \frac{2\zeta}{w_n}s + 1)} \tag{23}$$

where w_n is the natural frequency of the hydraulic cylinder piston, ζ is the damping ratio. K_{ce} is the total flow pressure coefficient. Above:

$$w_n = A_{me}\sqrt{\frac{4\beta_e}{V_e m}}, \zeta = \sqrt{\frac{\beta_e m}{V_e}} + \frac{B_p}{4A_{me}}\sqrt{\frac{V_e}{\beta_e m}}$$
$$K_{ce} = K_c + C_{tp}$$

Valve-controlled hydraulic motor

The slewing device of the excavator is an important part of motion control. In this paper, the transfer function of the valve-controlled hydraulic motor is as follows:

$$\theta_0 = \frac{\frac{K_q}{D_m}X_v - \frac{K_{ce}}{D_m^2}(1 + \frac{V_t}{4\beta_e K_{ce}}s)T_L}{s\left(\frac{s^2}{w_h^2} + \frac{2\zeta_h}{w_h^2}s + 1\right)} \tag{24}$$

where θ_0 is the rotation angle of the rotary motor, and T_L is the external load torque. ζ_h is the damping ratio coefficient, and D_m is the hydraulic motor displacement.

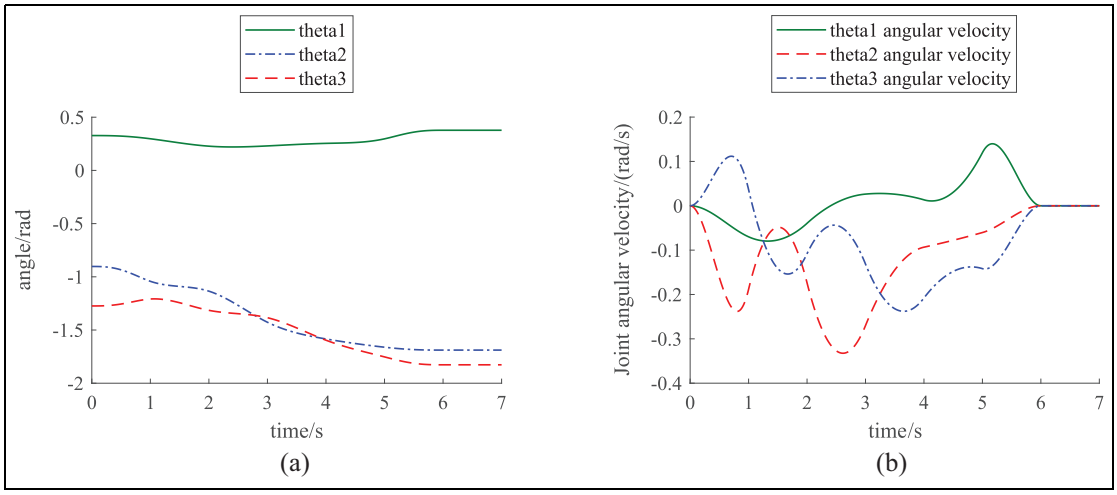


Figure 8. Joint angle planning. (a) Joint angle and (b) joint angular velocity.

Tracking control strategy and simulation analysis

Control objective

The trajectory of the autonomous excavator is planned based on the needs of the work.[28,29] This section defines the point-to-point trajectory path of the manipulator bucket's tooth.

In this section, the interpolation method is selected to calculate the trajectory planning. Eight key points of excavation are set in the excavation trajectory. To facilitate the subsequent planning of the excavation stage, the joint angle trajectory between each two points of the boom, arm, and bucket can be expressed by an interpolation polynomial. The angular velocity of the joint can be expressed by the first derivative of the polynomial, and the angular acceleration of the joint can be expressed by the second derivative of the polynomial. The general formula for trajectory planning of boom, arm, and bucket is established as follows:

$$\begin{cases} \theta_{i1}(t) = a_{10} + a_{11}t + a_{12}t^2 + a_{13}t^3 + a_{14}t^4 \\ \theta_{i2}(t) = a_{20} + a_{21}t + a_{22}t^2 + a_{23}t^3 \\ \theta_{i3}(t) = a_{30} + a_{31}t + a_{32}t^2 + a_{33}t^3 \\ \theta_{i4}(t) = a_{40} + a_{41}t + a_{42}t^2 + a_{43}t^3 \\ \theta_{i5}(t) = a_{50} + a_{51}t + a_{52}t^2 + a_{53}t^3 \\ \theta_{i6}(t) = a_{60} + a_{61}t + a_{62}t^2 + a_{63}t^3 \\ \theta_{i7}(t) = a_{70} + a_{71}t + a_{72}t^2 + a_{73}t^3 + a_{74}t^4 \end{cases} \quad (25)$$

where a_{mn} is the polynomial coefficient in the interpolation polynomial function. m, n are polynomial coefficients.$\theta_{ij}(t)$ the interpolation function of the j-segment of the i joint angle of the working device, which is the joint angle interpolation function of each segment in the 1-2-3-4-5-6-7-8 trajectory.

The optimization goal of this research is to lower the manipulator hydraulic cylinder movement stroke by reducing the change of the joint angle. The optimization objective function is as follows:

$$J = \min \sum_{i}^{3} \sum_{j}^{5} \left(\theta_{i,j+1} - \theta_{i,j}\right) \quad (26)$$

where i represents the i th joint angle. j represents the key point of the mining path at j.

The constraint function is shown in Formula (27):

$$\begin{aligned} \max\{|V_{ij}|\} &\leq V_{j\max} \\ \max\{|A_{ij}|\} &\leq A_{j\max} \end{aligned} \quad (27)$$

where V_{ij} is the angular velocity of the i th joint angle in the j-segment. A_{ij} is the joint angle acceleration of the i th joint angle in the j-segment; V_{jmax} and A_{jmax} represent the maximum angular velocity and angular acceleration limit of the corresponding joint angle, respectively.

For the modified autonomous excavator, the trajectory planning of the pose space is optimized and transformed into the joint angle of the joint space. The joint angle and angular velocity of a mining planning are shown in Figure 8.

Control strategy

The precise motion control strategy diagram of the manipulator of the autonomous excavator is shown in Figure 9. The traditional control strategy usually does not compensate for the external disturbance to the hydraulic system caused by the manipulator model itself in the dynamic model. In this paper, the dynamic online compensation method is used to compensate for

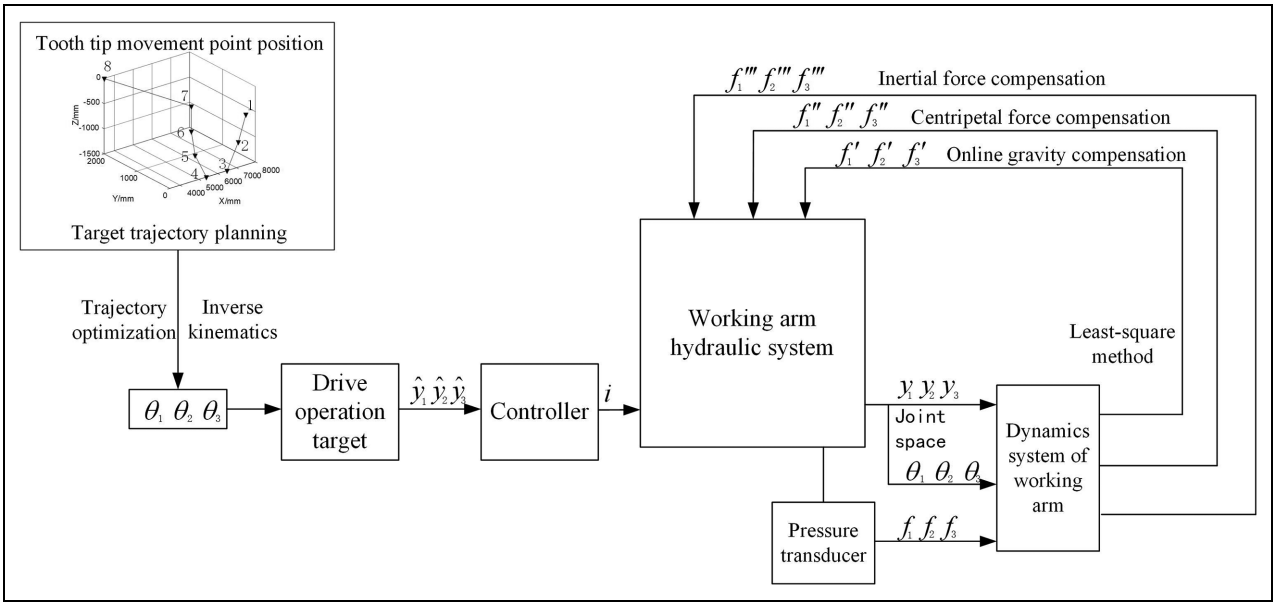


Figure 9. Precise motion control strategy of the manipulator of the autonomous excavator.

the external interference force in the direction of the hydraulic cylinder generated during the operation of the manipulator. It effectively eliminates the influence of external interference during the movement of the manipulator and improves the motion control accuracy of the excavator manipulator.

Define the tracking error of the system as:

$$\varepsilon = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \end{bmatrix} = \begin{bmatrix} \theta_{1d} - \theta_1 \\ \theta_{2d} - \theta_2 \\ \theta_{3d} - \theta_3 \end{bmatrix} \tag{28}$$

Define the sliding surface of the system as:

$$s = \begin{bmatrix} s_1 \\ s_2 \\ s_3 \end{bmatrix} = c\varepsilon + \dot{\varepsilon} = c \begin{bmatrix} \theta_{1d} - \theta_1 \\ \theta_{2d} - \theta_2 \\ \theta_{3d} - \theta_3 \end{bmatrix} + \begin{bmatrix} \dot{\theta}_{1d} - \dot{\theta}_1 \\ \dot{\theta}_{2d} - \dot{\theta}_2 \\ \dot{\theta}_{3d} - \dot{\theta}_3 \end{bmatrix} \tag{29}$$

Among them, $c = [\, c_1 \quad c_2 \quad c_3 \,], c_i > 0, i = 1, 2, 3$

Construct the Lyapunov function as:

$$V = \frac{1}{2}\left(s_1{}^2 + s_2{}^2 + s_3{}^2\right) \tag{30}$$

Set the control force to:

$$\tau = -M + M(\theta)\left[\ddot{\theta}_d + c(\theta_d - \theta) + ks + \eta \mathrm{sgn}(s)\right] \tag{31}$$

where M is the total interference torque of the system,

$$\theta_d = \begin{bmatrix} \theta_{1d} \\ \theta_{2d} \\ \theta_{3d} \end{bmatrix}, \theta = \begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \end{bmatrix}, \quad k, \eta \quad \text{Positive integer}$$

parameters for design, take the derivative of equation (30) and bring it into equation (31) to obtain.

$$\dot{V} = s^T \dot{s} = s^T \left[c(\theta_d - \theta) + \ddot{\theta}_d - \frac{1}{M(\theta)}(M + \tau) \right] \leqslant -ks^T s \leqslant 0 \tag{32}$$

It indicates that the designed control method is asymptotically stable, and the tracking error and its derivative approach 0 over time.

Simulation analysis

In the typical trapezoidal excavation environment, the simulation results of online compensation and without online compensation are compared using the developed autonomous excavator model as the control object. Figure 10 depicts the comparative simulation results with and without online compensation adjustment.

By substituting the trajectory planning path into the online compensation dynamic model, the gravity torque compensation, centripetal moment, and inertia moment generated during the movement of the manipulator can be obtained and the compensation amount is used in the system. Figure 11 depicts a time domain simulation of gravity moment compensation, centripetal moment compensation, and inertia moment compensation.

In the case of the reciprocating motion of the bucket hydraulic cylinder is stationary, the time domain curve of the driving force and displacement of the hydraulic cylinder are obtained by simulation as shown in Figure 12.

The excavator manipulator needs to overcome the external interference to work which consumes some energy and affects the trajectory tracking effect during

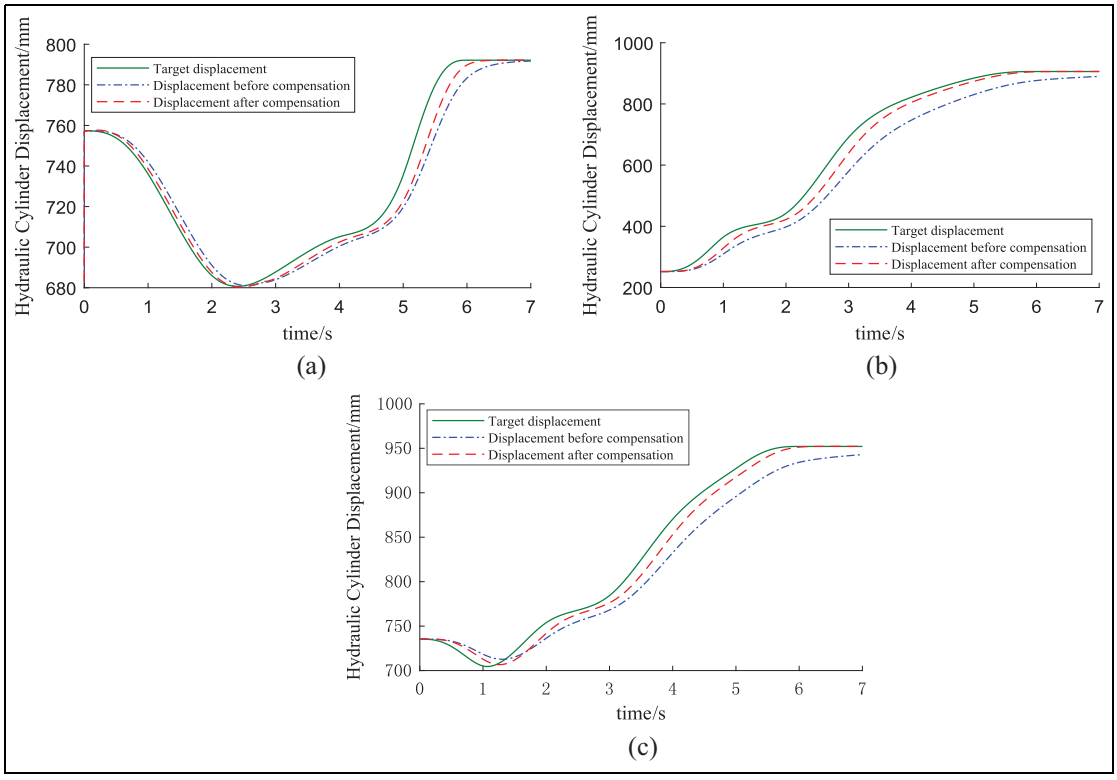


Figure 10. Displacement tracking curve of hydraulic cylinder. (a) Hydraulic cylinder displacement of the boom, (b) hydraulic cylinder displacement of the arm, and (c) hydraulic cylinder displacement of the bucket.

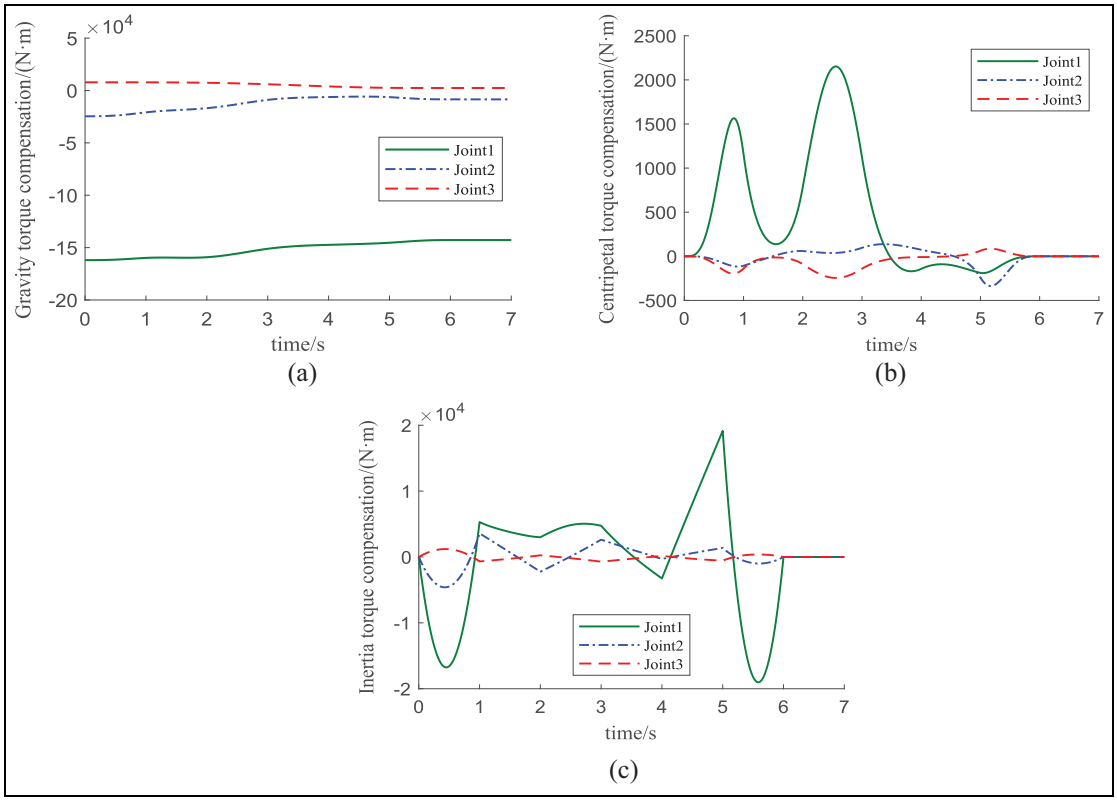


Figure 11. Online compensation of gravity, centripetal force, and inertia moment. (a) Gravity torque compensation, (b) centripetal force moment compensation, and (c) inertia moment compensation.

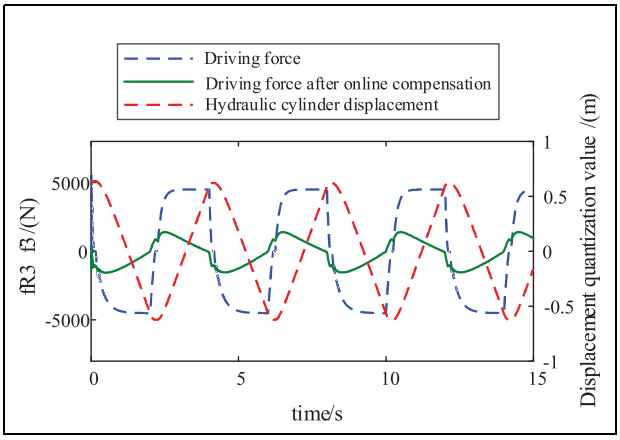


Figure 12. Bucket compensation result.

the operation process. The driving force after online compensation in the figure changes near 0. It can be seen that the online compensation control strategy effectively compensates for the inertia force, centripetal force, and gravity torque. The force change of the driving force enables the excavator manipulator to eliminate the influence of inertia, centripetal force, and gravity on the movement of the manipulator during the operation process, which can improve the working accuracy and efficiency of the autonomous operation of the excavator.

Test

The test equipment of the autonomous excavator is shown in Figure 13. The joint angle measurement units at the boom and arm are installed on their respective outer surfaces; the bucket uses a built-in displacement sensor, and the pressure sensor is installed at the inlet and outlet of the hydraulic cylinder pipeline. The bucket tooth tip space error is measured by the full transfer point method.

The four degrees of freedom of the manipulator and the rotary device of the autonomous excavator are driven by the hydraulic servo. To make the manipulator reach the target point accurately, the inclination sensor is installed at the rotation, boom, and arm shaft. A built-in displacement sensor is installed in the bucket hydraulic cylinder. During the operation process, the inclination sensor and the built-in length sensor transmit the collected data to the controller through the data bus, and obtain errors by comparing with the planned path of the joint angle, thereby achieving closed-loop control of the joint angle is carried out. The relevant component models are shown in Table 1.

In order to verify the effectiveness of the control strategy, the manipulator adopts the method of simulating tracking and excavating the trapezoidal path for experimental verification. Through the excavation trajectory planning introduced in the previous section, the

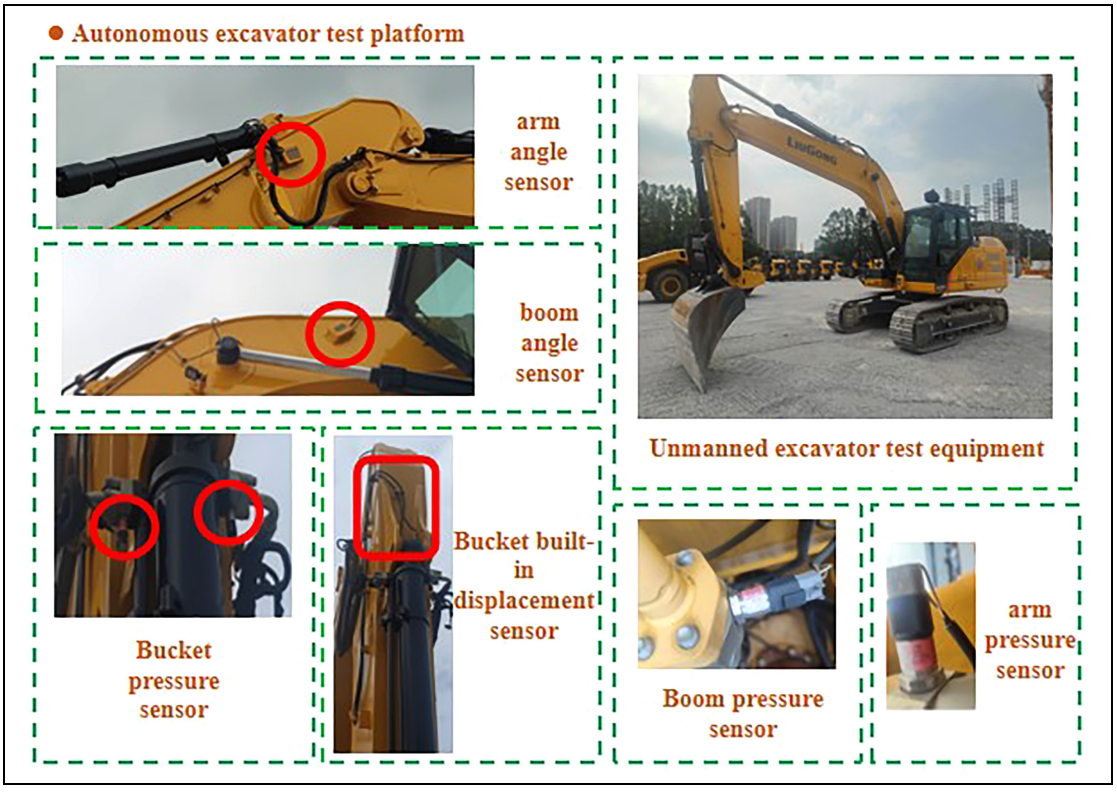


Figure 13. Autonomous excavator test platform.

Table 1. Related component parameters.

Sensor	Producer	Type
Tilt sensor	Pepperl + Fuchs Automation Engineering Co., Ltd.	INX360D-F99-I2E2-V15
Pressure transducer	Precision Electronics Limited	MEAS M5200
Hydraulic cylinder displacement sensor	Shenzhen Milan Technology Co., Ltd.	MTL3-1000MM-N/A-P

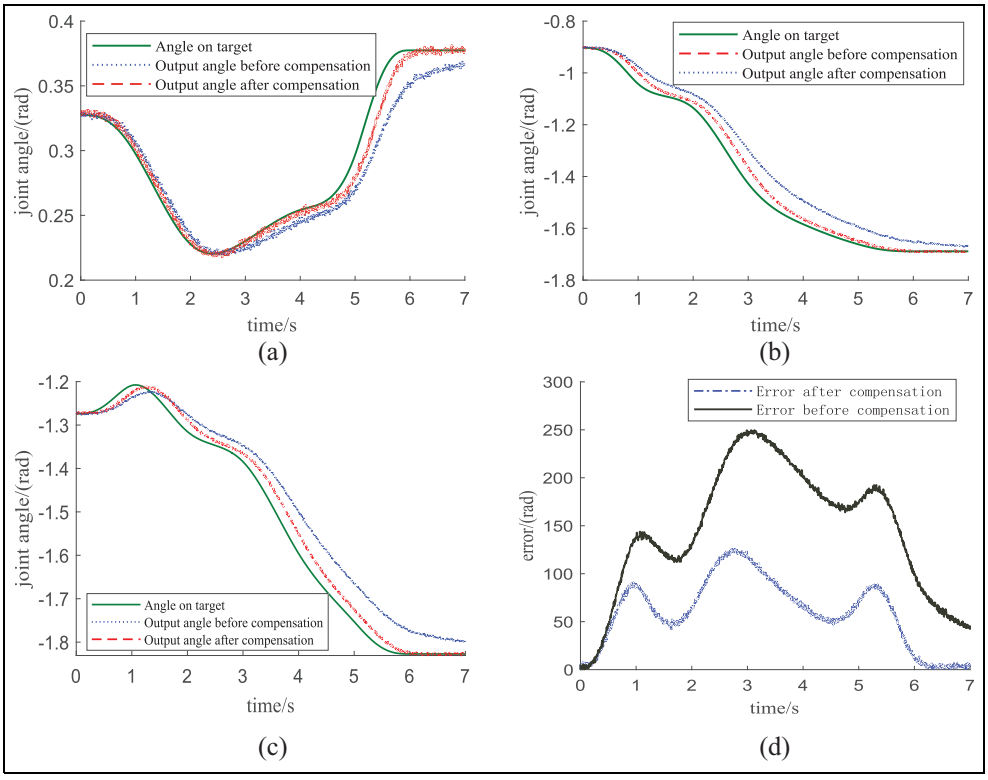


Figure 14. Tracking under load. (a) Boom joint angle, (b) arm joint angle, (c) bucket joint angle, and (d) bucket tooth tip tracking error analysis.

Table 2. Performance index comparison.

Hydraulic cylinder	Maximum tracking error/*rad*	Overall error/*rad*
Boom	0.04418	0.00429
	0.02512	0.00187
	(↓43.14%)	(↓56.41%)
Arm	0.08721	0.06529
	0.06405	0.02525
	(↓26.56%)	(↓61.33%)
Bucket	0.09639	0.03691
	0.04718	0.01319
	(↓51.05%)	(↓64.26%)

initial position of the manipulator is set according to the initial position of the excavation trajectory planning, so that the relevant joint angle is consistent with the planned initial joint angle, and then the tracking motion control is performed. In the case of bucket load, the online compensation control strategy is compared with the non-compensation control strategy. Using the above sensors to measure the joint angle and the hydraulic cylinder displacement feedback to the controller for the excavator arm tracking control. The test results are as follows:

It can be seen from Figure 14 that compared with the uncompensated manipulator control, the bucket tooth tip tracking accuracy after online compensation is significantly improved. Due to the large combined mass of the boom and its hydraulic cylinder and the load of the bucket, the tracking accuracy of the joint angle of the boom and the bucket increases obviously, and the tracking accuracy of the joint angle of the bucket rod is also greatly improved compared with that without compensation. The results are shown in Table 2.

It can be seen from Table 2 that under the load of the bucket, the maximum error of boom joint angle

tracking decreased by 43.14% respectively, the maximum error of arm joint angle and bucket joint angle tracking decreased by 26.56 and 51.05% respectively, and the average error of the whole trajectory are reduced by 56.41, 61.33, and 64.26% respectively, and the maximum error of bucket tooth tip space position tracking decreased by 50.42%. It can be seen that the online compensation control strategy and control strategy can effectively improve the control accuracy of the working device system of the autonomous operation excavator, and increase the operation accuracy and tracking performance of the manipulator of the autonomous operation excavator.

Conclusions

In this paper, the influence of gravity center, centripetal force, and inertial force on the control accuracy of the manipulator of the system autonomous excavator is studied by using the method of on-line compensation of external interference of hydraulic cylinder. The motion control strategy of the manipulator is proposed, and the simulation analysis and test are carried out. The main conclusions are as follows:

(1) The dynamic model of the autonomous excavator is established. The least square method is used to identify the dynamic model parameters of the excavator manipulator. The external interference of the bucket hydraulic cylinder is compensated online.
(2) The trajectory planning provides the trajectory control target of the autonomous operation excavator, and the autonomous operation test of the excavator is performed. The feasibility of the online compensation algorithm is verified, and the online compensation control strategy improves the autonomous operation accuracy of the excavator.
(3) The test results show that the online compensating control strategy improves the tracking performance of the autonomous excavator in manipulator self-tracking. This is of great significance to improve the tracking accuracy and increase the working efficiency of excavator autonomous operation.

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Appendix A

$$
J^1 = \begin{bmatrix} -BG_1 \cdot \sin(\theta_1 + \angle EBG_1) & 0 & 0 \\ BG_1 \cdot \cos(\theta_1 + \angle EBG_1) & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}
$$

$$
J^2 = \begin{bmatrix} -a_1 \sin\theta_1 - EG_2 \cdot \sin(\theta_{12} + \angle JEG_2) & -EG_2 \cdot \sin(\theta_{12} + \angle JEG_2) & 0 \\ a_1 \cos\theta_1 + EG_2 \cdot \cos(\theta_{12} + \angle JEG_2) & EG_2 \cdot \cos(\theta_{12} + \angle JEG_2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix}
$$

$$
J_v^1 = \begin{bmatrix} -BG_1 \cdot \sin(\theta_1 + \angle EBG_1) & 0 & 0 \\ BG_1 \cdot \cos(\theta_1 + \angle EBG_1) & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}
$$

$$
J_v^2 = \begin{bmatrix} -a_1 \sin\theta_1 - EG_2 \cdot \sin(\theta_{12} + \angle JEG_2) & -EG_2 \cdot \sin(\theta_{12} + \angle JEG_2) & 0 \\ a_1 \cos\theta_1 + EG_2 \cdot \cos(\theta_{12} + \angle JEG_2) & EG_2 \cdot \cos(\theta_{12} + \angle JEG_2) & 0 \\ 0 & 0 & 0 \end{bmatrix}
$$

$$
J_w^1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} J_w^2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix}
$$

$$
J^3 = \begin{bmatrix} -a_1 \sin\theta_1 - a_2 \sin\theta_{12} - JG_3 \cdot \sin(\theta_{123} + \angle LJG_3) & -a_2 \sin\theta_{12} - JG_3 \cdot \sin(\theta_{123} + \angle LJG_3) & -JG_3 \cdot \sin(\theta_{123} + \angle LJG_3) \\ a_1 \cos\theta_1 + a_2 \cos\theta_{12} + JG_3 \cdot \cos(\theta_{123} + \angle LJG_3) & a_2 \cos\theta_{12} + JG_3 \cdot \cos(\theta_{123} + \angle LJG_3) & JG_3 \cdot \cos(\theta_{123} + \angle LJG_3) \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}
$$

$$
J_v^3 = \begin{bmatrix} -a_1 \sin\theta_1 - a_2 \sin\theta_{12} - JG_3 \cdot \sin(\theta_{123} + \angle LJG_3) & -a_2 \sin\theta_{12} - JG_3 \cdot \sin(\theta_{123} + \angle LJG_3) & -JG_3 \cdot \sin(\theta_{123} + \angle LJG_3) \\ a_1 \cos\theta_1 + a_2 \cos\theta_{12} + JG_3 \cdot \cos(\theta_{123} + \angle LJG_3) & a_2 \cos\theta_{12} + JG_3 \cdot \cos(\theta_{123} + \angle LJG_3) & JG_3 \cdot \cos(\theta_{123} + \angle LJG_3) \\ 0 & 0 & 0 \end{bmatrix}
$$

$$
J_w^3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}
$$

(A – 1)

In the formula, $\theta_{12} = \theta_1 + \theta_2$, $\theta_{123} = \theta_1 + \theta_2 + \theta_3$.The above $\boldsymbol{J}^1$, $\boldsymbol{J}^2$, and $\boldsymbol{J}^3$ are the Jacobi matrix of the boom, the Jacobi matrix of the arm, and the Jacobi matrix of the bucket respectively.

$$
\begin{aligned}
M_{11} &= m_3[a_1^2 + a_2^2 + JG_3^2 + 2a_1a_2\cos\theta_2 \\
&\quad + 2\cdot a_1JG_3\cos(\theta_{23} + \angle LJG_3) + 2\cdot JG_3\cdot a_2\cos(\theta_3 + \angle LJG_3)] \\
&\quad + m_2\left[a_1^2 + EG_2^2 + 2\cdot a_1\cdot EG_2\cos(\theta_2 + \angle JEG_2)\right] \\
&\quad + m_1\cdot BG_1^2 + I_1 + I_2 + I_3 \\
M_{12} &= M_{21} = m_3[a_2^2 + JG_3^2 + a_1\cdot a_2\cos\theta_2 \\
&\quad + 2JG_3\cdot a_2\cos(\theta_3 + \angle LJG_3) + JG_3\cdot a_1\cos(\theta_{23} + \angle LJG_3)] \\
&\quad + m_2\left[EG_2^2 + a_1\cdot EG_2\cos(\theta_2 + \angle JEG_2)\right] + I_2 + I_3 \\
M_{13} &= M_{31} = m_3[JG_3^2 + JG_3\cdot a_2\cos(\theta_3 + \angle LJG_3) \\
&\quad + JG_3\cdot a_1\cos(\theta_{23} + \angle LJG_3)] + I_3 \\
M_{22} &= m_3\left[a_2^2 + JG_3^2 + 2\cdot JG_3\cdot a_2\cos(\theta_3 + \angle LJG_3)\right] \\
&\quad + m_2\cdot EG_2^2 + I_2 + I_3 \\
M_{23} &= M_{23} = I_3 + m_3[JG_3^2 + JG_3\cdot a_2\cos(\theta_3 + \angle LJG_3)] \\
M_{33} &= I_3 + m_3\cdot JG_3^2
\end{aligned}
\tag{A-2}
$$

In the formula, $\boldsymbol{I}_1$, $\boldsymbol{I}_2$, and $\boldsymbol{I}_3$ are the moment of inertia of the rigid body relative to G_1, G_2, and G_3 points. The details are as follows

$$
I_k = \begin{bmatrix}
\frac{-{}^kI_x + {}^kI_y + {}^kI_z}{2} & {}^kI_{xy} & {}^kI_{xz} & m_k{}^kx_{Gi} \\
{}^kI_{xy} & \frac{{}^kI_x - {}^kI_y + {}^kI_z}{2} & {}^kI_{yz} & m_k{}^ky_{Gi} \\
{}^kI_{xz} & {}^kI_{yz} & \frac{{}^kI_x + {}^kI_y - {}^kI_z}{2} & m_k{}^kz_{Gi} \\
m_k{}^kx_{Gi} & m_k{}^ky_{Gi} & m_k{}^kz_{Gi} & m_k
\end{bmatrix}
\tag{A-3}
$$

$$
\begin{aligned}
&C_{111} = C_{212} = C_{221} = C_{222} = C_{313} = C_{331} = C_{323} = C_{332} = C_{333} = 0 \\
&C_{112} = C_{121} = \dot\theta_1\dot\theta_2\{-m_3[a_1a_2\sin\theta_2 + JG_3a_1\sin(\theta_{23} + \angle LJG_3)] - m_2a_1EG_2\sin(\theta_2 + \angle JEG_3)\} \\
&C_{113} = C_{131} = \dot\theta_1\dot\theta_3\{-m_3[a_1JG_3\sin(\theta_{23} + \angle LJG_3) + JG_3a_2\sin(\theta_3 + \angle LJG_3)]\} \\
&C_{122} = \dot\theta_2\dot\theta_2\{-m_3[a_1a_2\sin\theta_2 + JG_3a_1\sin(\theta_{23} + \angle LJG_3)] - m_2a_1EG_2\sin(\theta_2 + \angle JEG_2)\} \\
&C_{123} = C_{132} = -\dot\theta_2\dot\theta_3m_3[JG_3a\sin(\theta_3 + \angle LJG_3) + JG_3a_1\sin(\theta_{23} + \angle LJG_3)] \\
&C_{133} = -\dot\theta_3\dot\theta_3m_3[JG_3a_2\sin(\theta_3 + \angle LJG_3) + JG_3a_1\sin(\theta_{23} + \angle LJG_3)] \\
&C_{211} = \dot\theta_1\dot\theta_1\{m_3[a_1a_2\sin\theta_2 + JG_3a_1\sin(\theta_{23} + \angle LJG_3)] - m_2a_1EG_2\sin(\theta_2 + \angle JEG_3)\} \\
&C_{213} = C_{231} = -\dot\theta_1\dot\theta_3m_3JG_3a_2\sin(\theta_3 + \angle LJG_3) \\
&C_{223} = C_{232} = -\dot\theta_2\dot\theta_3m_3JG_3a_2\sin(\theta_3 + \angle LJG_3) \\
&C_{233} = -\dot\theta_3\dot\theta_3m_3JG_3a_2\sin(\theta_3 + \angle LJG_3) \\
&C_{311} = \dot\theta_1\dot\theta_2m_3[JG_3a_2\sin(\theta_3 + \angle LJG_3) + JG_3a_1\sin(\theta_{23} + \angle LJG_3)] \\
&C_{312} = C_{321} = \dot\theta_1\dot\theta_2m_3JG_3a_2\sin(\theta_3 + \angle LJG_3) \\
&C_{322} = \dot\theta_2\dot\theta_2m_3JG_3a_2\sin(\theta_3 + \angle LJG_3)
\end{aligned}
\tag{A-4}
$$